# Rushil Rawat

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## Education

**University of Massachusetts, Amherst** B.S. Computer Science + Business
*Jr. CTO @ UMass Student Government · AI @ MIT · CodePath · Dean's List* *Aug 2024 – May 2028*

## Experience

**AI/ML Fellow** May 2025 – May 2026
*Break Through Tech @ MIT; Chambers Capital Ventures, Inc* *Remote*

- Engineered **NLP segmentation** on 10,000+ survey responses using **TF-IDF**, sentiment, and feature encoding
- Tuned **K-means** (k=2–8) to 0.58 silhouette, translating 8 behavioral clusters into 5 client-facing buyer personas
- Developed Streamlit dashboard mapping 11 purchase motivators to persona traits for startup marketing decisions

**Data Analytics Extern** Jun 2025 – Sep 2025
*Beats by Dre via Extern.com* *Remote*

- Analyzed 5,000+ customer responses with Python/**SQL**, VADER/**TextBlob** features on 6 audio trends
- Applied **scikit-learn** and **K-means** to uncover 5 customer groups and 4 loyalty indicators via correlation analysis
- Built **SQL ETL** workflows aggregating 12 metrics across 5 product lines into Tableau reports for market analysis

**Undergraduate Researcher** Dec 2024 – Feb 2025
*Wireless Sensors and System Lab, UMass Amherst* *Hybrid*

- Fine-tuned a ResNet CNN using **TensorFlow & OpenCV** for UAV fire detection (0.81 accuracy, 0.78 precision)
- Validated 2,000+ UAV image annotations with Python scripts and LabelImg, expanding dataset by 40%
- Integrated model inference into **GIS** dashboard and stress-tested low-visibility imagery, documenting failure modes

## Projects

**Garhwali Language Lab** | *PyTorch, Hugging Face, Transformers, LoRA, LangChain, Whisper* Sep 2026

- Published 2 Garhwali speech datasets on **Hugging Face**: 113K+ recordings (155 hrs), earning 1,500+ downloads
- Adapted **IndicBERTv2** with **LoRA** across 3 seeds, raising validation word accuracy from 24.3% to 28.3%
- Orchestrated resumable **LangChain** ingestion with overlap-free validation on 130GB data, gated by 832 CI tests

**HarmScope** | *Python, FAISS, HDBSCAN, Transformers, DuckDB, statsmodels, LLM/RAG* Aug 2026

- Architected **NLP** pipeline on 20M+ CFPB complaints by **embeddings**, FAISS to find emerging consumer harms
- Backtested time-series anomaly detection on 112 enforcement actions, cutting false alerts 10.7% to 0.45%
- Prototyped a grounded **LLM/RAG** layer with hybrid vector/keyword retrieval, citation-verified answers.

**CapitolWatch Insider Trading** | *Python, FastAPI, PostgreSQL, React, Docker* Jul 2026

- Designed **PostgreSQL** ETL pipeline for 72,000+ congressional trades using fuzzy name matching
- Evaluated trade timing of 350+ members with permutation testing and FDR correction, flagging 2 anomalies
- Reduced **ETL** pipeline time by 88% (4 hours to 29 mins) by content-hash **caching**, served via **FastAPI** and React

## Skills

**Languages**: Python, SQL, R, TypeScript/JavaScript, Java, C++
**AI & Machine Learning**: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, NLP, LLMs, RAG, LangGraph, LangChain, Fine-Tuning (LoRA), Embeddings, Vector Search (FAISS), OpenCV
**Backend & ML Systems**: FastAPI, REST APIs, Docker, Git, CI/CD, Linux, AWS, PostgreSQL, MySQL
**Data & Statistics**: Pandas, NumPy, SciPy, statsmodels, Model Evaluation, Statistical Testing, Anomaly Detection, ETL/Data Pipelines, Tableau